Leela

(3)

Molecular weight and size of polymers

Explain the term polydispersity.

Polydispersity:

Polydispersity is a parameter which gives an idea of the lowest and the highest molecular weight species as well as the distribution pattern of the intermediate molecular weight species.

By knowing $\bar{M}_w$ and $\bar{M}_n$ and then $\frac{\bar{M}_w}{\bar{M}_n}$, we can determine the extent of polydispersity. As the $\frac{\bar{M}_w}{\bar{M}_n}$ becomes higher, the molecular weight distribution curve becomes broader.

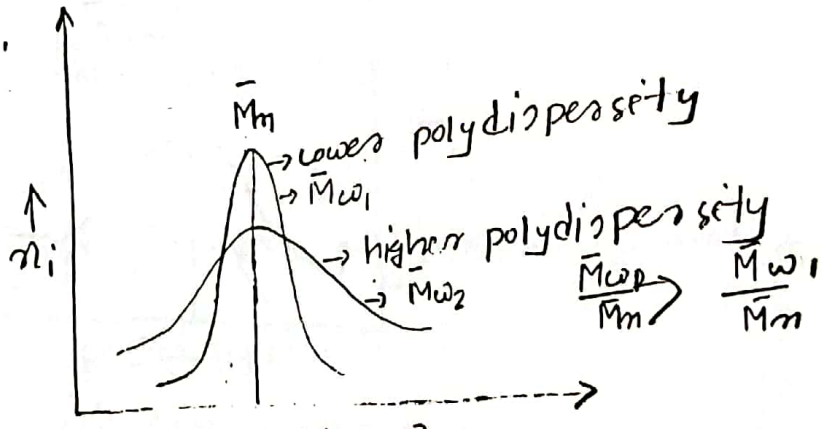


Fig: M.W distribution curve for two hypothetical polymer sample with different polydispersity

* what is meant by polydispersity index? what are the conditions for a monodispersed and polydisperse polymers?

Polydispersity index:

The ratio of the weight average molecular weight ($\bar{M}_w$) and the number average molecular weight ($\bar{M}_n$) is defined as the polydispersity index.

$$\therefore \text{Polydispersity index (PDI)} = \frac{\bar{M}_w}{\bar{M}_n}$$

It gives us an information about the distribution pattern of a given polymer.

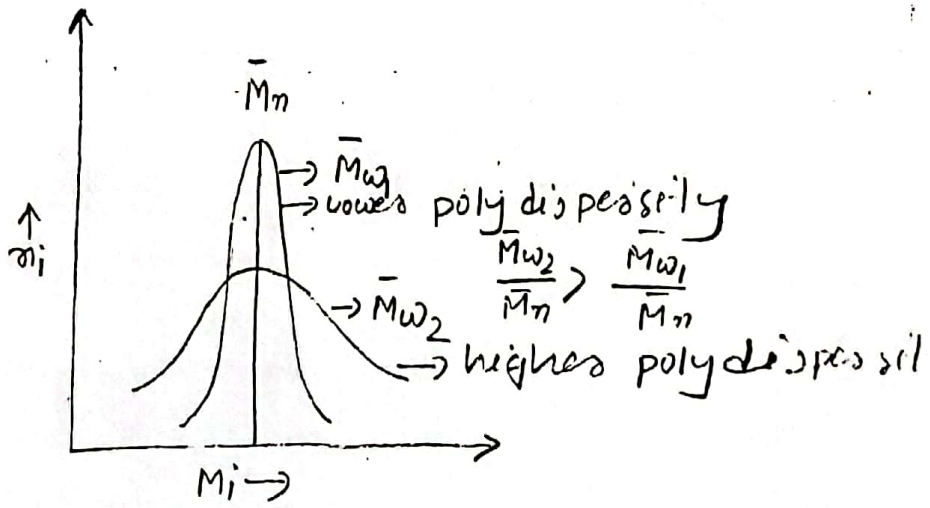


Fig: M.w distribution curve for two hypothetical sample with same M.w but different polydispersities.

For monodisperse polymers-

$$\frac{\bar{M}_w}{\bar{M}_n} = 1 \quad \text{or.} \quad \bar{M}_w = \bar{M}_n$$

For polydisperse polymers-

$$\frac{\bar{M}_w}{\bar{M}_n} > 1 \quad \text{or} \quad \bar{M}_w > \bar{M}_n$$

Draw and explain distribution of molecular weight in a typical polymer.

Distribution of molecular weight in a polymer:

The molecular weight distribution of polymers describe the relationship between the number of moles of each polymer species (N_i) and the molar mass (M_i) of that species.

For monodisperse polymers, all molecules have the same molecular weight.

Ex: water, alcohol.

whereas for polydisperse polymers, molecules are of different sizes. Thus have different molecular weight.

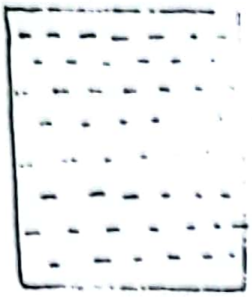


Fig: Monodispersed system

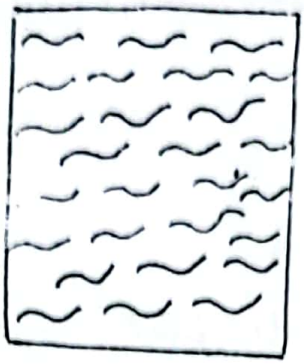


Fig: Polydispersed system

weight

Molecular distribution curve is computed by plotting against the number of fraction (n_i) of molecules against corresponding molecular weight (M_i).

M.w distribution curve

For a hypothetical polydisperse polymer sample-

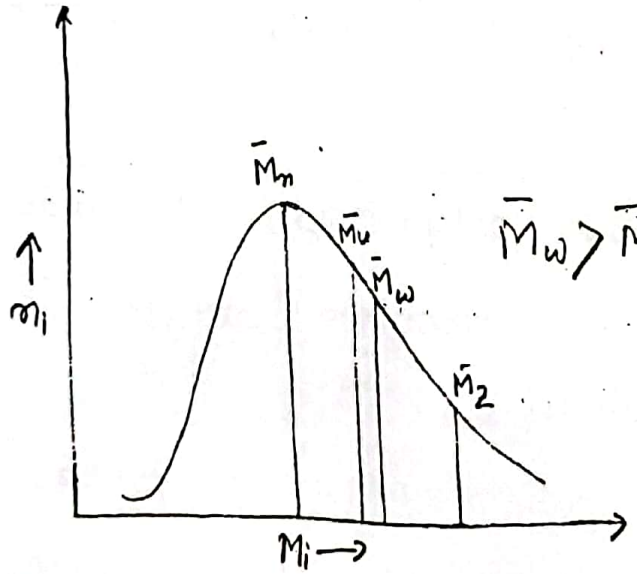


fig:- M.w distribution curve for a hypothetical polydispersed polymer sample.

(Fig: 1)
The curves brings out the relative positions of various average molecular weights in a polydisperse polymer sample.
Again. M.W distribution curve for two hypothetical polydisperse polymer sample—

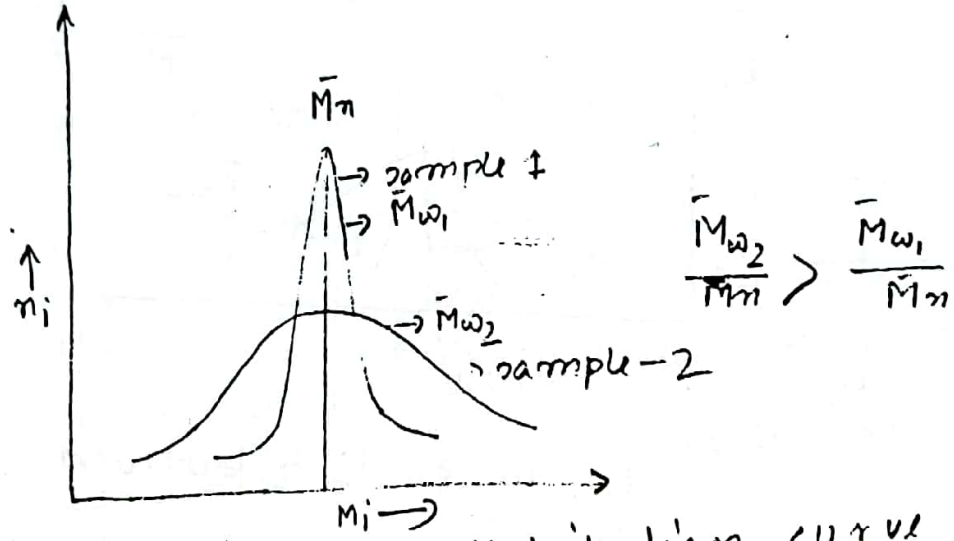


Fig-2: M.W distribution curve for two hypothetical polymer samples with different polydispersities.

This curve (Fig-2) shows the molecular weight distribution in two samples having the same number average molecular weight but different polydispersities. Here sample 1 has a narrower dispersion pattern and hence has a lower polydispersity than sample 2.

M.w distribution curve

similarly, for three hypothetical polymer sample-

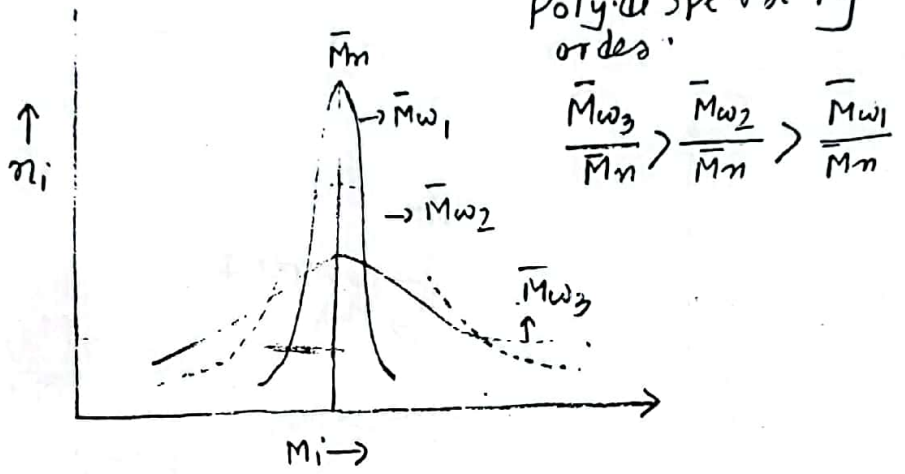


Fig: M.w distribution curve for three hypothetical samples with same M.w but different poly dispersities.

* Define the terms —

i) Number average molecular weight:

i) weight average molecular weight

i) viscosity average molecular weight.

ii) z-average molecular weight.

Number average molecular weight:

The number average molecular weight of a polymer is defined as the total weight of the sample divided by the number ~~of chains in the sample~~ of molecules present in the polymer chains of the sample..

If n_i is the number of molecules having molecular weight M_i, then the number average molecular weight $\bar{M}_n$ is given by -

$$\bar{M}_n = \frac{\Sigma n_i M_i}{\Sigma n_i}$$

Let, there are n number of molecules in a polymer sample.

Now, Total no. of molecules is given by -

$$n = n_1 + n_2 + n_3 + \cdots + n_i = \Sigma n_i$$

Number of molecules in fraction 1 = n_1

Number of fraction of fraction 1 = $\frac{n_1}{\Sigma n_i}$

M.W contribution by fraction 1 = $\frac{n_1 M_1}{\Sigma n_i}$

similarly,

M.w contribution by fraction 2 $= \frac{n_2 M_2}{\Sigma n_i}$

Therefore,

No. average molecular weight of whole polymer is given by-

$$\frac{n_1 M_1}{\Sigma n_i} + \frac{n_2 M_2}{\Sigma n_i} + \cdots\cdots + \frac{n_i M_i}{\Sigma n_i} = \frac{\Sigma n_i M_i}{\Sigma n_i} = \bar{M}_n$$

weight average molecular weight:

The weight average molecular weight is defined as

$$\bar{M}_w = \frac{\Sigma n_i M_i^2}{\Sigma n_i M_i}$$

let n number of molecules in a polymer sample.

Total weight of the polymer, $w = \Sigma n_i M_i$

weight of fraction 1 = $w_1 = n_1 M_1$

weight fraction of fraction 1 = $\frac{w_1}{w} = \frac{n_1 M_1}{\Sigma n_i M_i}$

M.w. contribution by fraction 1 = $\frac{n_1 M_1 \times M_1}{\Sigma n_i M_i} = \frac{n_1 M_1^2}{\Sigma n_i M_i}$

similarly,

M.w. contribution by fraction 2 = $\frac{n_2 M_2^2}{\Sigma n_i M_i}$

.....

Therefore, the weight average molecular weight of whole polymer -

$$\frac{n_1 M_1^2}{\Sigma n_i M_i} + \frac{n_2 M_2^2}{\Sigma n_i M_i} + \cdots + \frac{n_i M_i^2}{\Sigma n_i M_i} = \frac{\Sigma n_i M_i^2}{\Sigma n_i M_i} = \bar{M}_w$$

Viscosity average molecular weight:

The viscosity average molecular weight is defined as-

$$\bar{M}_v = \left(\frac{\Sigma n_i M_i^{1+a}}{\Sigma n_i M_i} \right)^{1/a}$$

where,

n_i = Number of molecules present in polymer chain.

M_i = Molecular weight of that polymer.

$\bar{M}_v$ = Viscosity average molecular weight.

'a' = A variable in the Mark-Houwink equation which relates intrinsic viscosity $[\eta]$ with the ~~visos~~ viscosity average molecular weight ($\bar{M}_v$).

z-average molecular weight:

The z-average molecular weight is defined as -

$$\bar{M}_z = \frac{\Sigma n_i M_i^3}{\Sigma n_i M_i^2}$$

where, n_i = No. of molecules present in polymer chain

M_i = M.w of that polymer.

$\bar{M}_z$ = z-average molecular weight

* calculate the number average molecular weight for a polystyrene for which the degree of polymerization is 26000.

we know —

$$\bar{M}_n = \bar{D}_p \cdot m$$

$$= 104.0\ g\,mol^{-1} \times 26000$$

$$= 2.704 \times 10^6\ g\,mol^{-1}$$

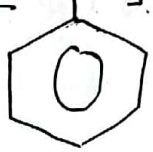
Ans:

where —

Degree of polymerization,

$\bar{D}_p = 26000$

M.w of monomer of polystyrene, $m = 104.00\ g/mol^{-1}$



Number average M.w, $\bar{M}_n = ?$

* Describe the light scattering method for the determination of molecular weight of a polymer.

Light scattering method:

The light scattering method ~~for the~~ is a convenient method for the determination of molecular weight of polymer. Light scattering method is based on the principle that when a light passed through a polymer molecule, the ~~original~~ incident light is scattered in different direction and also decrease its intensity."

The intensity measurement is used to determine the molecular weight of the polymer. The incident light on polymer is scattered in different directions as following-

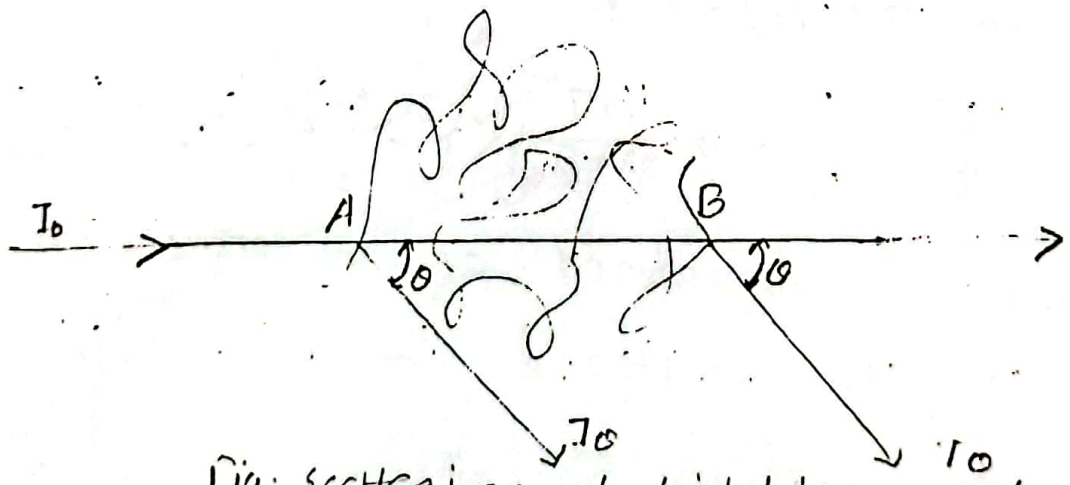


Fig: scattering of light by a polymer molecule.

Now, Debye develops a relation between molecular weight of the solute particle and the intensity of scattered light (whose particles of polymer solution is small compared to the wavelength) as following –

$$\frac{Kc}{R_\theta} = \frac{Hc}{\tau} = \frac{1}{\bar{M}_w P(\theta)} + 2Bc \quad \text{....(i)}$$

Again, particle scattering factor is related to wavelength as–

$$\frac{1}{P(\theta)} = 1 + \frac{16\pi^2}{3\lambda^2} (\bar{s}^2) \sin^2\left(\frac{\theta}{2}\right) \quad \text{....(ii)}$$

where – R_θ = Rayleigh ratio at observation angle

$\langle \bar{s}^2 \rangle$ = Mean square radius of gyration

c = concentration of solution

B = second medium co-efficient

K, H = Light scattering calibration constant.

P(θ) = Particle scattering factor

Now, substituting the value of $\frac{1}{P(\theta)}$ from (ii) in eqn (i) we get–

$$\frac{Kc}{R_\theta} = \frac{Hc}{\tau} = \frac{1}{\bar{M}_w} + \frac{16\pi^2}{3\lambda^2 \bar{M}_w} \langle \bar{s}^2 \rangle \sin^2\theta/2 + 2Bc \quad \text{....(iii)}$$

P(θ) can be comidered as a correction factor to the scattered intensity at

different values of θ.

such as-

$P(\theta) = 1$ at $\theta = 0$

similarly, $Bc = 0$ at $c = 0$

Now eqn (iii) becomes-

$$\left(\frac{Kc}{R_\theta}\right)_{c \to 0} = \frac{1}{\bar{M}_w} + \frac{16\pi^2}{3\lambda^2 \bar{M}_w} \langle \bar{s}^2 \rangle \sin^2(\theta/2) \quad \text{---- (iv)}$$

$$\Rightarrow \left(\frac{Kc}{R_\theta}\right)_{\theta \to 0} = \frac{1}{\bar{M}_w} + 2Bc \quad \text{---- (v)}$$

$$\left(\frac{Kc}{R_\theta}\right)_{c\theta \to 0} = \frac{1}{\bar{M}_w} \quad \text{---- (vi)}$$

From eqn (iv), (v), (vi), we can determine the molecular weight $\bar{M}_w$ of a polymer.

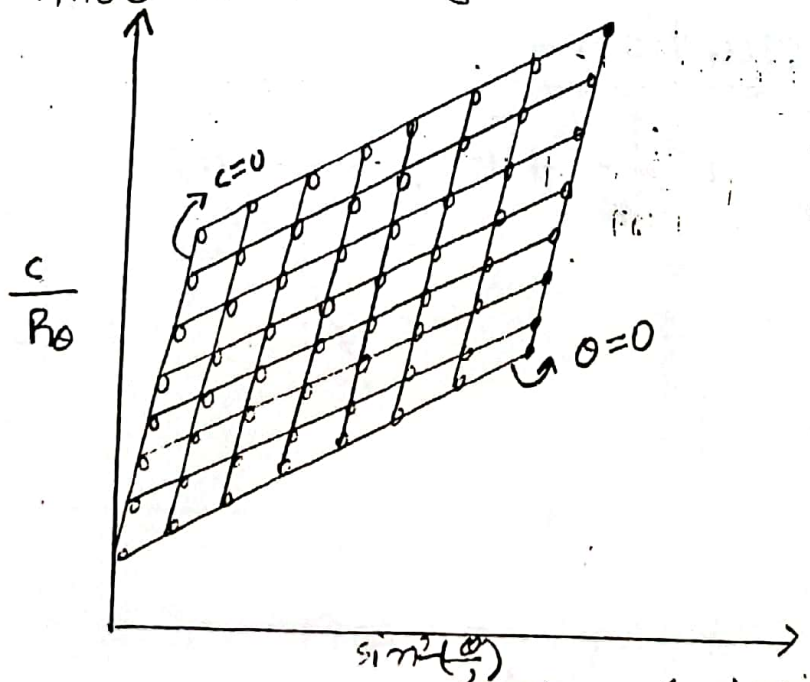


Fig: A typical zimm plot

* Describe the cryoscopic method for the determination of molecular weight of a polymers.

cryoscopic method:

The cryoscopic method for the determine of molecular weight of a polymer is based on the principle that the freezing point of a solvent is lowered when a non volatile solute is dissolved in it. Amount of depression can be expressed as follow

$$\Delta T_f = K_f m$$

K_f = cryoscopic constant
m = molal concentration

The depression of freezing point ΔT_f is related to the number average molecular weight and concentration through the following equation-

$$\frac{\Delta T_f}{c} = \frac{RT_f^2}{e\Delta H_f}\left(\frac{1}{\bar{M}_n} + Bc\right) \quad \text{---- (i)}$$

where,

ΔT_f = Depression of freezing point

c = concentration of the solution

ρ = density of the solvent

R = Molar gas constant

T_f = Freezing point of the solvent.

B = second virial co-efficient.

ΔH_f = Heat of fusion of the solvent.

From the freezing point depression curve (temperature vs time) for solvent and solution, ΔT_f value can be evaluated.

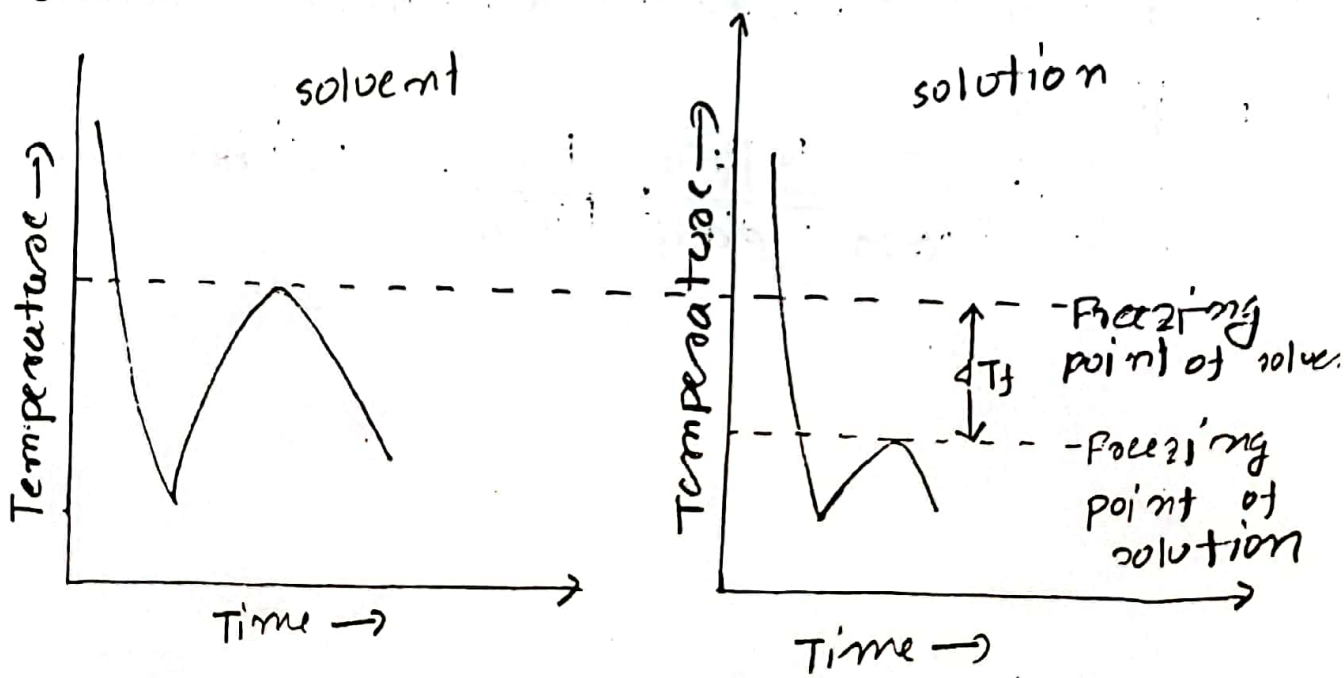
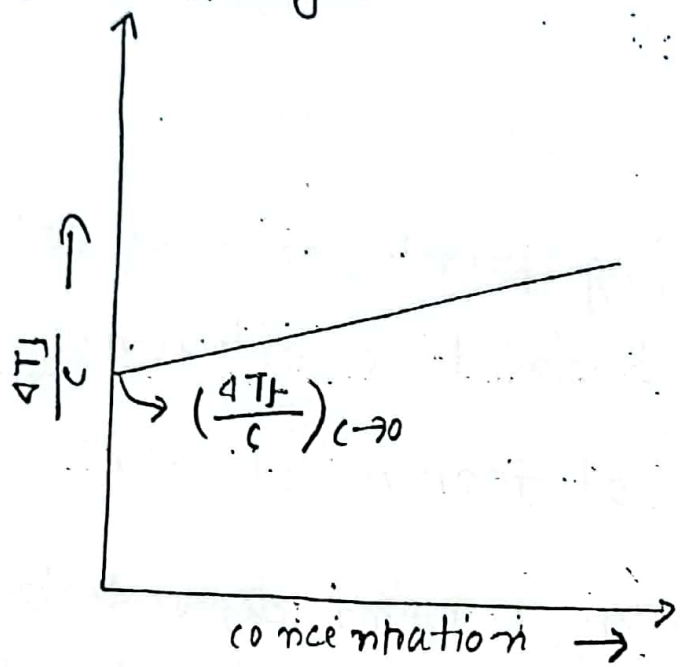


Fig: Freezing point curves for a solvent and a solution.

Now, a graph for $\frac{\Delta T_f}{C}$ versus concentration at infinite dilution i.e. zero concentration gives a straight line.



From the value of the ordinate intercept of the plot at zero concentration $\bar{M}_n$ can be determined from the equation—

$$\left(\frac{\Delta T_f}{C}\right)_{C \to 0} = \frac{R T_f^2}{\rho \Delta H_f} \cdot \frac{1}{\bar{M}_n} \quad \text{---(ii)}$$

* Describe membrane osmometric method for the determination of ~~use~~ molecular weight of a polymers.

Membrane osmometry:

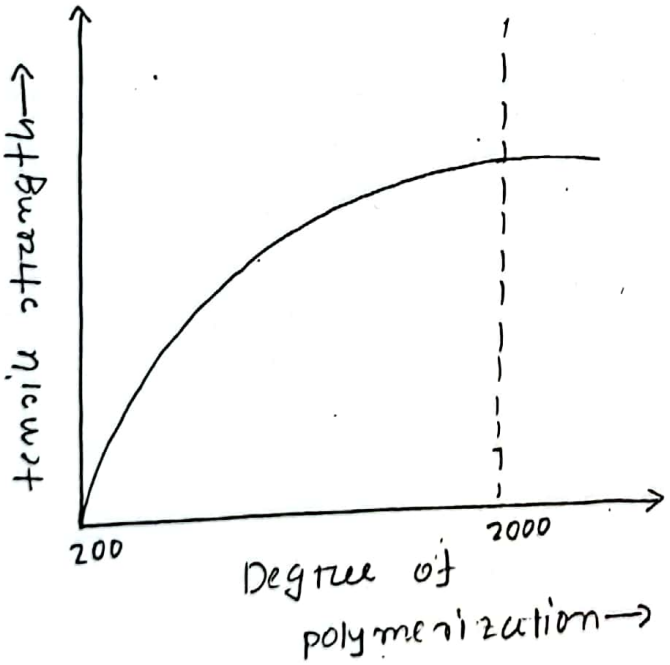
Membrane osmometry or osmotic pressure measurement method is a very widely used technique for the determination the number average molecular weight of polymers. It is based on the phenomena osmosis. "If a pure solvent is separated from the solution by a semipermeable membrane on account of the difference in chemical potentials.

* why it is important to learn about molecular weight of polymers?

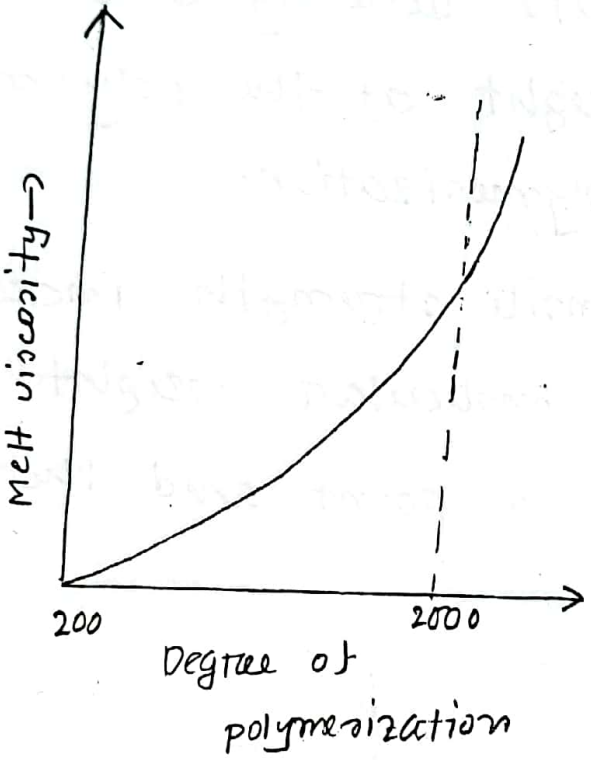
Many commercially useful polymers are selected on the bases of their properties such as melt viscosity, tensile strength.

These properties are directly dependent on the molecular weight of the polymer or its degree of polymerization.

For example, tensile strength increase with increasing molecular weight. The increase is upto a point and then it levels off.



The melt viscosity of the polymer however shows a different trend. At very high molecular weights, the melt viscosity rises more steeply than at lower molecular weights.



For this reason, it is important to learn about molecular weight of polymer.